

Deren Lei

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EDUCATION

University of California, Santa Barbara (UCSB)

SANTA BARBARA, CA

Computer Science B.S.

June 2019

College of Engineering Honors Program

GPA: 3.90

Planning to Declare: Minor in Statistical Science

RESEARCH EXPERIENCE

UCSB Natural Language Processing Lab

SANTA BARBARA, CA

Researcher (Advisor: William Wang)

Fall 2017 – Present

- **Stabilized deep reinforcement learning for dialogue (Initiator, work in progress)**
Deep reinforcement learning in NLP have problems of high variance predictions and sparse reward signals. Google's recently released BERT can be a good language modeling prior and solve the above issues.
 - **Inconsistency information solving (Coauthor, in submission)**
Recent entity typing tasks moves to more diverse and fine-grained datasets and current learning approaches have contradiction predictions. Improved performances by avoiding inconsistency outputs.
 - **Multimodal headline generation (Coauthor, in submission)**
Performed AI powered news content platform company generate attractive smartphone notifications based on contents. Collected currently one of the largest dataset from news: headline, story highlights, article, images, categories, etc to help model generates headline via different resources.
 - **Implicit regularization of stochastic gradient descent in NLP (First author, in submission)**
Studied the implicit regularization of stochastic gradient descent in practical deep learning systems and its correlation with explicit techniques. Results shed lights on the directions for future deep learning theory studies and also help researchers measure models performances better. ([arXiv:1811.00659](#) [cs.CL])
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WORK EXPERIENCE

Amazon

BEIJING, CHINA

Software Engineering Intern (Mentor: Gang Wang)

Summer 2017

- Created an internal tool to help developers to do automatic product page status checking. Saved them more than 5 hours of manual checking the special offer pages per month.
 - Performed more than 100000 data processing to test a softline product recommendation deep learning algorithm that could be applied to Amazon product recommendation system in recent years.
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ACADEMIC PROJECTS

Activity Recognition (Advisor: B.S. Manjunath)

Winter 2018

- Helped researchers in UCSB Vision Research Lab measuring Residual Computer Networks' performances by learning via single image frames on AVA dataset using ResNet101. The results can help their project of analyzing people's behavior in hospital.

NoteHub (Developer, Product Manager)

Winter 2018

- An software system with android-end and web-end (mainly focused) for students and professors to share notes in both hand-written and electrical format using Java and Javascript.
- Contacted Scanbot Team to freely use their API for education purpose for a quarter when it costs €100,000.

GaUCHoMap (Group Leader)

Spring 2017

- Developed a map based event sharing system on android platform using Google Maps APIs to help UCSB students involved in social activities. Received hands-on android development experience with seniors while learning basic engineering skills with other sophomores.
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SKILLS

Programming Languages: Python, Java, C++, Scala, MATLAB, R, MIPS Assembly, Javascript

Technical Tools: Tensorflow, PyTorch, scikit-learn, Git, AWS, LaTeX, SQL, Firebase